

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_no_replay.
- Best TSPLIB holdout gap: tsplib_no_replay.
- Best synthetic holdout gap: tsplib_no_replay.

Run Metadata

- run_name: run_20260513_123823_a
- started_at_local: 2026-05-13 12:38:23
- finished_at_local: 2026-05-13 12:50:40
- duration_hhmm: 00:12
- duration_seconds: 736.891
- seed_offset: 0
- replicate_label: a
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.113341	0.0	0.072126	0.743278	0.76	0.03087
tsplib_random_replay	random	score_only	False	0.145818	0.0	0.092793	0.669863	0.76	0.058627
tsplib_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.903546	0.56	-0.027564
tsplib_failure_replay_compression	failure	novelty_gate	True	0.174803	0.016511	0.117242	0.0	0.76	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.113341, synthetic 0.0, combined 0.072126.

- Accepted-epoch count 4, mean accepted code novelty 0.743278, and final complexity 0.76.
- Adaptation efficiency 0.03087 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.113341 across 7 instances; family means: ch=0.131838, kroD=0.106368, pcb=0.211312, pr=0.157592, rd=0.036662, st=0.017778.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3393, "family": "a", "name": "a280", "optimality_gap": 0.315626}, {"best_known_cost": 629, "cost": 778, "family": "eil", "name": "eil101", "optimality_gap": 0.236884}, {"best_known_cost": 14379, "cost": 17673, "family": "lin", "name": "lin105", "optimality_gap": 0.229084}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.145818, synthetic 0.0, combined 0.092793.
- Accepted-epoch count 2, mean accepted code novelty 0.669863, and final complexity 0.76.
- Adaptation efficiency 0.058627 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.145818 across 7 instances; family means: ch=0.121798, kroD=0.192777, pcb=0.30013, pr=0.025657, rd=0.165234, st=0.093333.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3470, "family": "a", "name": "a280", "optimality_gap": 0.345483}, {"best_known_cost": 14379, "cost": 17603, "family": "lin", "name": "lin105", "optimality_gap": 0.224216}, {"best_known_cost": 7542, "cost": 8535, "family": "berlin", "name": "berlin52", "optimality_gap": 0.131663}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.903546, and final complexity 0.56.
- Adaptation efficiency -0.027564 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.174803, synthetic 0.016511, combined 0.117242.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.174803 across 7 instances; family means: ch=0.140367, kroD=0.200902, pcb=0.239139, pr=0.301186, rd=0.112769, st=0.088889.
- Panel synthetic_holdout mean gap 0.016511 across 4 instances; family means: clustered_gaussian=0.024674, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 20172, "family": "lin", "name": "lin105", "optimality_gap": 0.402879}, {"best_known_cost": 2579, "cost": 3434, "family": "a", "name": "a280", "optimality_gap": 0.331524}, {"best_known_cost": 7542, "cost": 9619, "family": "berlin", "name": "berlin52", "optimality_gap": 0.275391}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Results

Condition	Replay Mode	Selection Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (Mean)	Complexity	Interpretation
tsplib_no_replay	None	Score Only	**0.1133**	0.0	**0.0721**	0.743	0.76	Best optimality gaps and transfer metrics; no replay yielded the best transfer despite high code novelty.
tsplib_random_replay	Random Replay	Score Only	0.1458	0.0	0.0928	0.67	0.76	Lower code novelty than no replay but worse transfer and TSPLIB gaps; replay did not improve transfer.
tsplib_failure_replay	Failure Replay	Score Only	0.2134	0.0649	0.1594	0.90	0.56	Highest gaps in all metrics despite highest code novelty; failure replay negatively impacted transfer and optimality.
tsplib_failure_replay_compression	Failure Replay + Compression	Novelty Gate	0.1748	0.0165	0.1172	0.0	0.76	Code novelty dropped to zero while transfer and TSPLIB gaps worsened vs. no replay; compression pressure appears detrimental for transfer.

Key Interpretations

- ****Transfer Metrics Priority****: According to the main evidence (held-out TSPLIB and synthetic gap), ****tsplib_no_replay**** outperforms all replay conditions for transfer and optimality.
- ****Replay Impact****: Replay modes (random, failure) result in higher mean TSPLIB optimality gaps and transfer gaps compared to no replay.
- ****Code Novelty vs. Transfer****:
 - Failure replay condition increases code novelty but correlates with worse transfer and optimality.
 - Compression-aware failure replay reduces code novelty to zero but does not improve transfer compared to random or failure replay without compression.
 - Thus, increases in code novelty do not imply algorithmic invention or better transfer.
- ****Replay Modes Differences****:
 - No replay yields better transfer and optimization results.

- Random replay degrades transfer slightly with lower code novelty than no replay.
- Failure replay leads to highest gaps and code novelty, indicating instability or less effective adaptation.
- Compression-aware failure replay further reduces code novelty but fails to improve transfer, indicating compression pressure may hinder performance in replay.

Conservative Conclusion

The ****no replay**** condition is optimal for transfer and held-out performance. Replay (random or failure) lowers transfer quality despite varying effects on code novelty. Failure replay conditions produce higher optimality gaps and worse transfer metrics, and compression pressure reduces code novelty without performance gains. Hence, replay modes tested do not improve generalization or transfer relative to no replay. Also, code novelty metrics should not be conflated with algorithmic invention absent improvements in transfer or optimality gaps.